



Counter Artificial Intelligence - A Preliminary Analysis of the Capabilities of AI Models on Local Systems

J. Wolffrom

jwsc@ruc.dk

Supervisor:

Hemant Ghayvat

hghayvat@ruc.dk

May 17, 2026

Roskilde University

Department of Computer Science

Abstract

Contents

1	Glossary	3
2	Disclosure	3
3	Introduction	3
3.1	The Problem	4
3.1.1	Research Question	4
3.2	Scope and Limitations	4
4	Linux Commandline	6
4.1	Processes and Jobs	6
4.1.1	Process Architecture and Concepts	6
4.1.2	Pipelines and Process Communication	7
4.1.3	Job Control and Management	8
4.1.4	Process Monitoring and Discovery	8
4.1.5	Process Control and Termination	9
4.1.6	Process Scheduling and Priority Management	9
4.1.7	Process Dependencies and Synchronization	10
4.2	Filesystems	10
4.2.1	Blocks, Clusters and Sectors	11
4.2.2	Filesystems and The Virtual File System	11
4.2.3	Inodes	11
4.2.4	Partitions	11
4.2.5	LEFTOVERS	12
4.3	User Management and Groups	14
5	Analysis	15
6	Discussion	16
7	Conclusion	17
8	Future Work	18
9	References	19
A	Linux Commands Reference	20

1 Glossary

list of terms and definitions:

- **Process** - An instance of a program that is being executed. It contains the program code and its current activity.
- **Pipeline** - A series of data processing stages where the output of one stage is the input to the next.
- **Thread** - An isolated unit of execution that shares memory space with other threads of the same process.
- **Job** - A process or a pipeline that is started or scheduled to run by the user in a shell session or from the terminal.

2 Disclosure

Any and all resources related to- and used in this study can be found here:

<https://github.com/X1las/CAI>

Everything in this paper has been written by human hands and proofread by Claude 4.5 Sonnet.

AI has been used for spell checking, sentence structure corrections, as well as text reductions, however everything has been checked by a human to make sure edits do not conflict with the cited theory.

The list of commands in appendix ?? was sourced from SS64 n.d., and compiled/categorized by the previously mentioned model. This is not considered a significant contribution to the paper, and is not cited as a source for any of the analysis or conclusions. It is simply a reference for readers who may be interested in the commands that were considered for the simulation design.

3 Introduction

In August 2025, Anthropic’s Claude model was reportedly used in one of the first documented cases of “vibe hacking” as outlined in Alex Moix, J. K., Ken Lebedev. (2025). *Threat intelligence report: August 2025*. Anthropic. In that case, the attacker used the model to collect publicly available vulnerability information, identify organizations using the affected software, and exploit the vulnerability to gain access to internal infrastructure. The incident was notable because it required limited human involvement, while the model assisted in producing malicious tooling used for extortion.

If AI continues to develop at the current pace, similar incidents are likely to become more frequent. It is also plausible that some future attacks will be increasingly autonomous, if not fully. In that context, the need for stronger cybersecurity capabilities will grow, and many experts argue that AI will play a central role in both attack and defense strategies. This motivates research into how AI models behave when they share the same system and are tasked with detecting, disrupting, or countering each other.

3.1 The Problem

The aim of this research paper is not to predict exactly how AI models would counteract each other, but to explore how they might use available system tools in a simulated local environment, and whether useful insights can be gained from this process.

In simple terms, the problem is twofold. First, it is unclear which terminal commands are realistically available to an AI agent running locally with broad system access. Second, it is not known how an AI model should be trained to use such commands effectively, or which combinations are most useful for detecting and counteracting another model in the same environment.

3.1.1 Research Question

The research question is then as follows:

How can AI agents detect and counteract each other through Linux filesystem operations and process management commands in a local system environment?

- **1.** What Linux filesystem and process management commands (including `ps`, `kill`, `fuser`, file operations, and permission management) can be used by an AI agent to monitor, identify, and interact with another AI agent on the same system?
- **2.** How can filesystem-based communication mechanisms (including file creation, modification, symbolic links, and access control) serve as both attack vectors and detection methods in AI-to-AI interactions?
- **3.** What strategies emerge when two AI agents are simulated in a Linux environment with access to process monitoring, job scheduling, and file manipulation commands, and what patterns of behavior can be identified?
- **4.** What are the implications of filesystem-centric AI defense mechanisms for the future of autonomous cybersecurity systems?

To address these questions, we will first compile a list of terminal commands on linux that can be used to outline a base of actions for the simulation. This will be limited to linux for the sake of simplicity, but the process can be expanded to include Windows commands as well.

3.2 Scope and Limitations

This research is intentionally scoped to focus on the Linux operating system and its associated filesystem operations. All commands, system calls, utilities, and code examples presented in this paper relate specifically to the Linux environment and its filesystem-centric architecture.

This limitation has several implications. First, the commands and techniques discussed are primarily applicable to Linux-based systems and may not directly translate to other operating systems such as Windows or macOS, which have fundamentally different filesystem structures, permission models, and command-line interfaces. Second, while Linux systems support various network operations and inter-process communication mechanisms, this study focuses on filesystem-based interactions, as these represent the most funda-

mental and universally available means of system manipulation and inter-agent communication.

The rationale for this scope is threefold: (1) Linux provides a well-documented, standardized command-line environment that is widely used in server and embedded systems where autonomous AI agents are most likely to operate; (2) the filesystem serves as a common substrate through which processes can interact, persist data, and observe system state, making it a natural focus for studying AI-to-AI interactions; and (3) limiting the scope to a single operating system allows for deeper analysis of specific commands and techniques rather than superficial coverage of multiple platforms.

Consequently, this paper does not address platform-specific considerations for Windows (e.g., PowerShell, Registry manipulation, NTFS permissions) or macOS (e.g., launchd, Keychain access, APFS features). Additionally, while network-based attacks and defenses are acknowledged as important vectors in real-world scenarios, they fall outside the scope of this filesystem-focused investigation.

4 Linux Commandline

A list of the linux commands used in this paper is included in Appendix A. This list is not an exhaustive list of all commands available on linux based distributions, it is based on the commands found on SS64 (n.d.), which includes most if not all the built-in bash commands. For a full list of linux commands, see Kerrisk (n.d.).

To begin with, we can derive that commands on the linux commandline can affect most of anything on the system. Processes, files, network connections, users and even hardware can be affected by commands. This means that in order to understand what can be done locally, we need to understand how the system works at its most fundamental level and what commands gives way to manipulate.

4.1 Processes and Jobs

Processes are a fundamental unit of execution in the Linux operating system, the management of which oversees all actions taken on the system and by the system. Understanding how these processes work, how they can be monitored and affected by the user, commands, and other processes is crucial in understanding how AI agents could operate in our simulated environment, and how they could potentially interact with each other.

For references on processes we refer to our previous work *Asynchronous Synergy: A study of performance in Multi-Core Threading across programming languages* (Wolffrom 2025), this report will not cover the finer details of process or thread management, but rather what they mean for the commandline.

4.1.1 Process Architecture and Concepts

There are four important keywords in this category, which are *processes*, *threads*, *pipelines* and *jobs*. Processes can be somewhat loosely defined as we discovered in our prior study (Wolffrom 2025), however they are generally considered to be instances of code that are running in an isolated memory space. Each process has its own virtual address space, preventing it from directly accessing the memory of other processes and this isolation is enforced by the kernel and provides fundamental security boundaries between processes.

When a process is created through the `fork()` system call, it receives a unique Process ID (PID). The newly created child process is initially an exact copy of the parent process, including its memory space, open file descriptors, and execution context. The child can then use `exec()` to replace its memory space with a new program. This fork-exec model is fundamental to how shells launch commands and how processes spawn subprocesses.

Threads can be generally split into two categories, virtual threads (also known as user-level threads) and kernel threads, where the former are managed by the process itself and the latter are managed by the operating system's scheduler, which is usually EEVDF (Community 2025, EEVDF Scheduler) in the linux environment. Processes are usually scheduled to run on kernel threads, and the kernel threads are scheduled to have equal time allotted on the CPU. Multiple threads within a single process share the same memory space, making inter-thread communication faster but also introducing synchronization challenges.

Processes maintain several important attributes beyond their PID, including a Parent

Process ID (PPID) that identifies which process created them, a user ID (UID) and group ID (GID) that determine privileges, and various resource limits. When a parent process terminates before its children, those child processes are typically reparented to the init process (PID 1), becoming orphaned processes. Conversely, when a child terminates but the parent has not yet read its exit status, it becomes a zombie process, retaining an entry in the process table until the parent collects its status.

4.1.2 Pipelines and Process Communication

A pipeline in the context of the linux commandline refers to the mechanism of connecting the output of one process to the input of another process using the `|` operator (“Piping in Unix or Linux” 2024). This is not related to the concept of pipelines in the context of data processing or machine learning, but it is a powerful tool for chaining commands together and creating complex workflows. An example of this would be the command `ps aux | grep python`, which would list all the processes running on the system and then filter the results to show only those that contain the word “python”. This allows for a user to easily search through the processes and find specific ones that they are interested in.

Pipelines are implemented using anonymous pipes, which are unidirectional data channels managed by the kernel. When a shell creates a pipeline, it forks multiple child processes and connects their standard output and input streams. The processes in a pipeline execute concurrently, with data flowing between them as it becomes available. All processes in a pipeline share the same process group ID (PGID), which allows them to be managed collectively.

Beyond anonymous pipes, Linux provides several other inter-process communication (IPC) mechanisms. Named pipes (FIFOs) persist in the filesystem and can be used by unrelated processes. Shared memory allows multiple processes to access the same memory regions for high-performance data exchange. Message queues provide structured communication channels. Sockets enable communication both locally and across networks. Understanding these mechanisms is crucial because they represent potential channels through which AI agents might exchange information or detect each other’s presence.

4.1.3 Job Control and Management

By cross referencing “Cron Command in Linux with Examples” (2026), Free Software Foundation (2025, May), “Job Scheduling Commands in Linux” (2025), and “Shell Scripting - JOB SPEC & Command” (2025), jobs are a concept in the linux commandline manuals that refer to processes and pipelines that are either started or scheduled to run by the user in a shell session or from the terminal. They are given a process ID (PID) on creation, regardless of whether they are multiple commands tied together in a pipeline or a single command. They can be run in the foreground or background, and they can be managed using built-in commands such as `jobs`, `fg`, `bg`, and `kill`.

Foreground jobs receive keyboard input and prevent the shell from accepting new commands until they complete or are suspended. Background jobs, launched by appending `&` to a command, run concurrently with the shell, allowing the user to continue issuing commands. The `Ctrl-Z` key combination suspends a foreground job, which can then be resumed in the background with `bg` or returned to the foreground with `fg`.

Job control is mediated through signals, which are asynchronous notifications sent to processes. Common signals include `SIGTERM` (polite termination request), `SIGKILL` (immediate forced termination), `SIGSTOP` (suspend execution), and `SIGCONT` (resume execution). The `kill` command, despite its name, is the primary interface for sending arbitrary signals to processes.

Using `cron`, a user can also schedule jobs to run at specific times or intervals, which can be useful for automating tasks and running processes without user intervention. The `crontab` command manages per-user cron schedules, while system-wide cron jobs are defined in `/etc/crontab` and `/etc/cron.d/`. The `at` and `batch` commands provide one-time scheduled execution. These scheduling mechanisms could allow an AI agent to persist its presence or execute actions at predetermined times.

Where processes usually differ from jobs is that they can be triggered and scheduled by the operating system without user involvement. Jobs are generally more manageable for users, as they are directly tied to the shell and can be easily manipulated using built-in commands. Processes, on the other hand, can be more complex and may require more advanced tools for management and monitoring. Due to their nature as well, processes can be more anonymous and harder to track as well, seeing as they can be triggered by other processes and may not have a direct association with a user or shell session.

4.1.4 Process Monitoring and Discovery

Commands such as `ps`, `top`, `htop` and `pgrep` can be used to monitor and search through processes, whilst commands such as `pkill`, `killall` in addition to `kill` all allow for termination of processes.

The `ps` command provides a snapshot of current processes. With options like `aux`, it displays all processes system-wide with detailed information including CPU and memory usage, execution time, and the command line used to launch each process. The `-ef` option provides a full-format listing showing parent-child relationships through PPID values. For AI agents, examining process command lines and parent relationships could reveal the presence of competing agents or suspicious activity.

The `top` and `htop` commands provide real-time, continuously updating views of system

processes, sorted by various metrics such as CPU or memory consumption. These interactive tools highlight processes that are consuming unusual amounts of resources, which could indicate malicious behavior or intensive computational tasks associated with an AI agent.

The **pgrep** command searches for processes by name or other attributes, returning only the PIDs. Its counterpart **pkill** sends signals to matching processes. The **pidof** command finds PIDs by exact command name. The **pstree** command displays the process hierarchy as a tree, making parent-child relationships visually apparent.

More sophisticated process introspection is available through the **/proc** pseudo-filesystem, where each process has a directory named by its PID containing files with detailed information: **/proc/[pid]/cmdline** shows the command line, **/proc/[pid]/exe** is a symlink to the executable, **/proc/[pid]/environ** contains environment variables, and **/proc/[pid]/fd/** lists open file descriptors. An AI agent with filesystem access could extensively analyze other processes through this interface.

4.1.5 Process Control and Termination

Beyond the basic **kill** command, Linux provides several mechanisms for process termination. The **killall** command sends signals to all processes matching a given name, while **pkill** offers more flexible pattern matching and attribute-based selection. The **timeout** command can be used to automatically terminate a process if it exceeds a specified execution time.

Process resource limits can be queried and modified using **ulimit** (for the current shell and its children) or **prlimit** (for arbitrary processes). These limits cover maximum file sizes, number of open files, memory usage, CPU time, and other resources. An AI agent could potentially use these mechanisms to constrain a competing agent's capabilities or to detect constraints imposed upon itself.

The **strace** command traces system calls made by a process, revealing every interaction with the kernel including file operations, network activity, and process management. Similarly, **ltrace** traces library calls. These powerful debugging tools could allow one AI agent to monitor another's behavior in detail, though they require appropriate privileges.

4.1.6 Process Scheduling and Priority Management

In addition to this, the EEVDF scheduler (Community 2025, EEVDF) allows for scheduling affinity using the **sched_setattr()** system call, which the user can call on directly on starting a process using the **nice** command, or after the process has started with **renice**. This allows for a user to give a process higher or lower priority on the CPU, which can make some processes run with more CPU time than others and vice versa. This can be used to give a process more resources to run, or to starve a process of resources and make it run slower.

Process priorities in Linux are expressed as "nice" values ranging from -20 (highest priority) to 19 (lowest priority), with 0 being the default. Despite the name, higher nice values are "nicer" to other processes by yielding more CPU time to them. Only privileged users can decrease nice values (increase priority), though any user can increase nice values (decrease priority) for their own processes.

Beyond nice values, Linux supports multiple scheduling classes. The default `SCHED_OTHER` (or `SCHED_NORMAL`) uses the `EEVDF` algorithm for time-sharing. Real-time scheduling classes (`SCHED_FIFO` and `SCHED_RR`) provide deterministic scheduling for time-critical applications but require elevated privileges. The `chrt` command can query and modify scheduling policies and priorities.

CPU affinity, controlled by `taskset`, restricts a process to execute only on specified CPU cores. This can be used for performance optimization but could also be weaponized to isolate or constrain competing processes. The `cgroups` (control groups) mechanism provides comprehensive resource management, allowing fine-grained control over CPU time, memory allocation, I/O bandwidth, and other resources for groups of processes.

4.1.7 Process Dependencies and Synchronization

In addition to these, there are also commands such as `wait` which can be used to pause the execution of a process until another process has finished, and `fuser` which can be used to identify which processes are using a specific file or socket. This can be useful for identifying which processes are interacting with each other and potentially interfering with each other.

The `wait` command is primarily used in shell scripts to synchronize parent processes with their children. Without `wait`, a parent might terminate before its children complete, or might not know whether child processes succeeded or failed. The command can wait for specific PIDs or for all background jobs.

The `fuser` command identifies processes using specific files or sockets, returning PIDs and access types (read, write, execute, or root directory). This is particularly valuable for detecting filesystem-based interactions between processes. The related `lsof` (list open files) command provides comprehensive information about all files opened by processes, including regular files, directories, network sockets, pipes, and devices. Since everything in Unix-like systems is treated as a file, `lsof` offers a powerful view into process behavior and inter-process relationships.

File locking mechanisms, accessible through the `flock` command or programmatically through system calls, allow processes to coordinate access to shared resources. An AI agent could use file locks both for legitimate synchronization and for detecting or blocking competing agents attempting to access the same resources.

4.2 Filesystems

Filesystems are the organizational structures that determine how data is stored and retrieved on a storage device. They define the way files are named, stored, and accessed, as well as how metadata is managed. Common filesystems in Linux include `ext4`, `XFS`, `Btrfs`, and `F2FS`, each with its own features and performance characteristics.

The Linux filesystem provides the fundamental infrastructure through which processes interact with persistent data and communicate with each other. Understanding file operations is crucial for analyzing how AI agents might manipulate system state, hide information, or detect the presence of other agents.

4.2.1 Blocks, Clusters and Sectors

Sectors refer to the smallest physical unit of storage on a traditional disk drive, typically 512 bytes of storage. Sectors used to be fixed in size due to the limitations and design of hard disk drives, but with modern solid-state drives the notion of sectors has become a logical abstraction rather than a physical constraint.

When a collection of sectors is grouped together, it forms a block/cluster depending on the filesystem. A block is a logical unit of storage that the filesystem uses to manage files. The size of blocks can vary depending on the filesystem and the underlying storage device, but common block sizes are 4KB or 8KB. Clusters are essentially the same as blocks, but they are often used in the context of NTFS filesystems that windows computers use.

Blocks are tracked by the filesystem's metadata structures, such as inodes or the superblock, which record which blocks are allocated to which files. When a file is created or modified, the filesystem allocates blocks to store the file's data and updates the metadata accordingly. The block size can impact performance and storage efficiency; larger blocks can reduce fragmentation but may waste space for small files, while smaller blocks can be more efficient for small files but may lead to increased fragmentation.

Blocks are the smallest logical unit of storage, meaning that even if a file is only a few bytes in size, it will still occupy at least one block on the filesystem. This can lead to inefficiencies, especially for small files, as the unused space within the block cannot be utilized by other files.

4.2.2 Filesystems and The Virtual File System

The Virtual File System (or VFS for short) is an abstraction layer within the Linux kernel that provides a uniform interface for different filesystem implementations. It allows the kernel to support multiple filesystem types (such as ext4, XFS, Btrfs, etc.) without requiring changes to user-space applications. The VFS manages file operations, directory structures, and metadata in a way that is consistent across all supported filesystems.

4.2.3 Inodes

Inodes (index nodes) are data structures used by many Unix-like filesystems to represent files and directories. Each inode contains metadata about a file, including its size, permissions, ownership, timestamps (creation, modification, access), and pointers to the data blocks where the file's content is stored. The inode does not contain the filename itself; instead, directory entries map filenames to their corresponding inode numbers.

When a file is accessed, the filesystem uses the directory structure to resolve the filename to an inode number, then retrieves the inode to access the file's metadata and data blocks. This separation of filename and file metadata allows for features like hard links, where multiple directory entries can point to the same inode, effectively giving a single file multiple names.

4.2.4 Partitions

Partitions are divisions of a storage device that allow it to be managed separately. Each partition can contain a different filesystem, enabling multiple operating systems or different types of data to coexist on the same physical device. Partitions are defined in a

partition table, which is typically located at the beginning of the storage device. Common partitioning schemes include MBR (Master Boot Record) and GPT (GUID Partition Table).

4.2.5 LEFTOVERS

Linux organizes storage into three conceptual levels: the virtual file system (VFS)(Richard Gooch n.d.), individual filesystems, and files themselves. The VFS provides a unified interface that abstracts the underlying filesystem implementations, allowing different filesystem types to coexist and be accessed through a common set of system calls.

At the top level, Linux employs a single hierarchical directory tree(“Unix File System” 2026) rooted at `/`, unlike Windows which uses drive letters. All storage devices, regardless of their physical location or filesystem type, are mounted into this unified tree structure. Common mount points include `/home` for user directories, `/tmp` for temporary files, and `/proc` or `/sys` for kernel-provided pseudo-filesystems that expose process and system information.

Individual filesystems can be of various types, each with distinct characteristics. Modern Linux distributions commonly use `ext4`, which provides journaling for crash recovery and supports large files and volumes. Alternative filesystems include `XFS` (optimized for parallel I/O operations), `Btrfs` (supporting snapshots and copy-on-write), and `F2FS` (optimized for flash storage). The choice of filesystem affects performance characteristics, maximum file sizes, and available features, though the VFS layer ensures that basic operations remain consistent across types.

Files can be created, modified, copied, moved, and deleted using a variety of commands. The `touch` command creates empty files or updates timestamps, while `mkdir` creates directories. File content can be written using redirection operators (`>` and `>>`) or commands like `echo`, `cat`, or text editors.

Copying and moving files is accomplished through `cp` and `mv` respectively. The `cp` command creates a new file with duplicated content, while `mv` relocates the file within the filesystem hierarchy without duplicating data when the source and destination are on the same filesystem. Deletion is performed with `rm` for files and `rmdir` or `rm -r` for directories.

More sophisticated operations include `rsync`, which can synchronize files between directories or systems while minimizing data transfer through checksums and delta encoding, and `dd`, which performs low-level block-by-block copying useful for creating disk images or manipulating raw devices.

At the filesystem implementation level, files are represented by inodes (index nodes), which are data structures containing metadata about files including permissions, ownership, timestamps, and pointers to the actual data blocks. The inode does not store the filename itself; instead, directory entries map filenames to inode numbers.

This separation enables the concept of links. A hard link is an additional directory entry pointing to the same inode, effectively giving a single file multiple names. All hard links are equivalent, and the file’s data is only deleted when the last link is removed and no processes have the file open. Hard links cannot span filesystems and cannot reference directories (to prevent circular references).

Symbolic links (symlinks), created with `ln -s`, are special files containing a path to another file. Unlike hard links, symlinks can reference directories and cross filesystem boundaries, but they break if the target is moved or deleted. The `readlink` command can resolve symbolic links to their targets.

File descriptors are integer handles that processes use to reference open files. When a process opens a file using the `open()` system call, the kernel returns a file descriptor that the process uses for subsequent operations. Standard file descriptors include 0 (stdin), 1 (stdout), and 2 (stderr). Commands like `ls -l` (list open files) can display which file descriptors are in use by which processes, making it possible to detect which files an agent might be accessing.

Linux implements a permission model that controls read, write, and execute access for three categories: the file's owner, members of the file's group, and all other users. Permissions are displayed by `ls -l` in the format `rw-rw-rw-`, where each trio represents owner, group, and other permissions respectively.

The `chmod` command modifies permissions using either symbolic notation (`chmod u+x file`) or octal notation (`chmod 755 file`). Ownership is changed with `chown` for the owner and `chgrp` for the group. The special permissions bits include the setuid bit (executes with owner's privileges), setgid bit (inherits directory's group), and sticky bit (restricts deletion in shared directories like `/tmp`).

Beyond traditional Unix permissions, modern Linux systems support Access Control Lists (ACLs), which provide finer-grained control by allowing permissions to be specified for arbitrary users and groups. ACLs are managed through `getfacl` and `setfacl` commands. Additionally, Security-Enhanced Linux (SELinux) and AppArmor provide mandatory access control frameworks that can restrict processes regardless of traditional file permissions.

File attributes, accessible through `lsattr` and modifiable via `chattr`, provide additional control mechanisms. For example, the immutable attribute (`+i`) prevents any modifications to a file, even by root, until the attribute is removed.

Filesystems must be mounted before they can be accessed, using the `mount` command which attaches a filesystem to a directory mount point. The `umount` command detaches filesystems. The `/etc/fstab` file specifies which filesystems should be automatically mounted at boot time.

Creating new filesystems is accomplished with the `mkfs` family of commands (e.g., `mkfs.ext4`, `mkfs.xfs`), while filesystem consistency checks and repairs are performed by `fsck`. These operations typically require elevated privileges and unmounted filesystems.

Monitoring filesystem usage is essential for detecting unusual behavior. The `df` command displays disk space usage by filesystem, while `du` shows space used by directories and files. The `stat` command provides detailed information about a specific file, including inode number, size, blocks allocated, and all timestamps (access time, modification time, and change time).

For detecting file system activity, several commands prove useful. The `find` command can search for files based on numerous criteria including name patterns, modification times, permissions, and size. The `inotify` system provides efficient filesystem event monitoring,

accessible through tools like `inotifywait`, which can watch for file creation, modification, deletion, and access in real time.

4.3 User Management and Groups

5 Analysis

6 Discussion

7 Conclusion

8 Future Work

9 References

References

- Alex Moix, J. K., Ken Lebedev. (2025). *Threat intelligence report: August 2025*. Anthropic.
- Community, T. K. D. (2025). *The linux kernel documentation - scheduler*. <https://docs.kernel.org/scheduler>
- Cron command in linux with examples. (2026). <https://www.geeksforgeeks.org/linux-unix/cron-command-in-linux-with-examples/>
- Free Software Foundation, I. (2025, May). *Gnu bash manual* (5.3). https://www.gnu.org/software/bash/manual/html_node/index.html#SEC_Contents
- Job scheduling commands in linux. (2025). <https://www.geeksforgeeks.org/linux-unix/job-scheduling-commands-in-linux/>
- Kerrisk, M. (n.d.). *Linux manual pages: Alphabetical list of all pages*. https://man7.org/linux/man-pages/dir_all_alphabetic.html
- Piping in unix or linux. (2024). <https://www.geeksforgeeks.org/linux-unix/piping-in-unix-or-linux/>
- Richard Gooch, P. E. (n.d.). *The linux kernel docs - virtual file system*. The Kernel Development Community. <https://docs.kernel.org/filesystems/vfs.html>
- Shell scripting - job spec & command. (2025). <https://www.geeksforgeeks.org/linux-unix/shell-scripting-job-spec-command/>
- SS64. (n.d.). *An a-z index of the linux command line: Bash + utilities*. <https://ss64.com/bash/>
- Unix file system. (2026). <https://www.geeksforgeeks.org/operating-systems/unix-file-system/>
- Wolffrom, J. (2025). *Asynchronous synergy: A study of performance in multi-core threading across programming languages*. University of Roskilde. <https://github.com/X1las/AsSynergy>

A Linux Commands Reference

The following is a complete A–Z index of Linux commands and utilities, sourced from <https://ss64.com/bash/>. Commands marked with • are Bash built-ins; all others are external utilities or Core Utils.

Command	Description
&	Start a new process in the background
alias •	Create an alias
apropos	Search Help manual pages (man -k)
apt	Search for and install software packages (Debian/Ubuntu)
apt-get	Search for and install software packages (Debian/Ubuntu)
aptitude	Search for and install software packages (Debian/Ubuntu)
aspell	Spell checker
at	Schedule a command to run once at a particular time
awk	Find and replace text, database sort/validate/index
basename	Strip directory and suffix from filenames
base32	Base32 encode/decode data and print to standard output
base64	Base64 encode/decode data and print to standard output
bash	GNU Bourne-Again SHell
bc	Arbitrary precision calculator language
bg •	Send to background
bind •	Set or display readline key and function bindings
break •	Exit from a loop
builtin	Run a shell builtin
bzip2	Compress or decompress named file(s)
cal	Display a calendar
caller •	Return the context of any active subroutine call
case	Conditionally perform a command
cat	Concatenate and print (display) the content of files
cd •	Change directory
cfdisk	Partition table manipulator for Linux
chattr	Change file attributes on a Linux file system
chgrp	Change group ownership
chmod	Change access permissions
chown	Change file owner and group
chpasswd	Update passwords in batch mode
chroot	Run a command with a different root directory
chkconfig	System services (runlevel)
cksum	Print CRC checksum and byte counts
clear	Clear the terminal screen/console (ncurses)
clear_console	Clear the terminal screen/console (bash)
cmp	Compare two files
comm	Compare two sorted files line by line
command •	Run a command - ignoring shell functions
continue •	Resume the next iteration of a loop
cp	Copy one or more files to another location
cpio	Copy files to and from archives
cron	Daemon to execute scheduled commands
crontab	Schedule a command to run at a later time
csplit	Split a file into context-determined pieces

Command	Description
curl	Transfer data from or to a server
cut	Divide a file into several parts
date	Display or change the date & time
dc	Desk calculator
dd	Data duplicator — convert and copy a file, write disk headers, boot records
ddrescue	Data recovery tool
ddrescueelog	Tool for ddrescue logfiles
declare •	Declare variables and give them attributes
df	Display free disk space
diff	Display the differences between two files
diff3	Show differences among three files
dig	DNS lookup
dir	Briefly list directory contents
dircolors	Colour setup for 'ls'
dirname	Convert a full pathname to just a path
dirs •	Display list of remembered directories
dos2unix	Windows/MAC to UNIX text file format converter
dmesg	Print kernel & driver messages
dpkg	Package manager (Debian/Ubuntu)
du	Estimate file space usage
echo •	Display message on screen
egrep	Search file(s) for lines that match an extended expression
eject	Eject removable media
enable •	Enable and disable builtin shell commands
env	Environment variables
ethtool	Ethernet card settings
eval •	Evaluate several commands/arguments
exec •	Execute a command
exit •	Exit the shell
expand	Convert tabs to spaces
export •	Set an environment variable
expr	Evaluate expressions
false •	Do nothing, unsuccessfully
fdformat	Low-level format a floppy disk
fdisk	Partition table manipulator for Linux
fg •	Send job to foreground
fgrep	Search file(s) for lines that match a fixed string
file	Determine file type
find	Search for files that meet a desired criteria
fmt	Reformat paragraph text
fold	Wrap text to fit a specified width
for	Expand words, and execute commands
format	Format disks or tapes
free	Display memory usage
fsck	File system consistency check and repair
ftp	File Transfer Protocol
function	Define function macros
fuser	Identify/kill the process that is accessing a file
gawk	Find and replace text within file(s)
getopts •	Parse positional parameters

Command	Description
getfacl	Get file access control lists
grep	Search file(s) for lines that match a given pattern
groupadd	Add a user security group
groupdel	Delete a group
groupmod	Modify a group
groups	Print group names a user is in
gzip	Compress or decompress named file(s)
hash •	Remember the full pathname of a name argument
head	Output the first part of file(s)
help •	Display help for a built-in command
history •	Command history
hostname	Print or set system name
htop	Interactive process viewer
iconv	Convert the character set of a file
id	Print user and group id's
if	Conditionally perform a command
ifconfig	Configure a network interface
ifdown	Stop a network interface
ifup	Start a network interface up
import	Capture an X server screen and save the image to file
install	Copy files and set attributes
iostat	Report CPU and i/o statistics
ip	Routing, devices and tunnels
jobs •	List active jobs
join	Join lines on a common field
kill •	Kill a process by specifying its PID
killall	Kill processes by name
klist	List cached Kerberos tickets
less	Display output one screen at a time
let •	Perform arithmetic on shell variables
link	Create a link to a file
ln	Create a symbolic link to a file
local •	Create a function variable
locate	Find files
login	Login to the computer
logname	Print current login name
logout •	Exit a login shell
look	Display lines beginning with a given string
lpc	Line printer control program
lpr	Print files
lprint	Print a file
lprintd	Delete a print job
lprintq	List the print queue
lprm	Remove jobs from the print queue
lsattr	List file attributes on a Linux second extended file system
lsblk	List block devices
ls	List information about file(s)
lsuf	List open files
lspci	List all PCI devices
make	Recompile a group of programs

Command	Description
man	Help manual
mapfile •	Read lines from standard input into an indexed array variable
md5sum	Compute and check MD5 message digest
mkdir	Create new folder(s)
mkfifo	Make FIFOs (named pipes)
mkfile	Make a file
mkisofs	Create a hybrid ISO9660/JOLIET/HFS filesystem
mknod	Make block or character special files
mktemp	Make a temporary file
modprobe	Add or remove modules from the Linux Kernel
more	Display output one screen at a time
most	Browse or page through a text file
mount	Mount a file system
mttools	Manipulate MS-DOS files
mtr	Network diagnostics (traceroute/ping)
mv	Move or rename files or directories
mmv	Mass move and rename (files)
nc	Netcat, read and write data across networks
netstat	Networking connections/stats
nft	nftables for packet filtering and classification
nice	Set the priority of a command or job
nl	Number lines and write files
nohup	Run a command immune to hangups
notify-send	Send desktop notifications
nslookup	Query Internet name servers interactively
op	Operator access
open	Open a file in its default application
openssl	OpenSSL command line
passwd	Modify a user password
paste	Merge lines of files
pathchk	Check file name portability
Perf	Performance analysis tools for Linux
ping	Test a network connection
pgrep	List processes by name
pkill	Kill processes by name
popd •	Restore the previous value of the current directory
pr	Prepare files for printing
printcap	Printer capability database
printenv	Print environment variables
printf •	Format and print data
ps	Process status
pushd •	Save and then change the current directory
pv	Monitor the progress of data through a pipe
pwd •	Print working directory
quota	Display disk usage and limits
quotacheck	Scan a file system for disk usage
ram	ram disk device
rar	Archive files with compression
rcp	Copy files between two machines
read •	Read a line from standard input

Command	Description
<code>readarray •</code>	Read from STDIN into an array variable
<code>readonly •</code>	Mark variables/functions as readonly
<code>reboot</code>	Reboot the system
<code>rename</code>	Rename files
<code>renice</code>	Alter priority of running processes
<code>remsync</code>	Synchronize remote files via email
<code>return •</code>	Exit a shell function
<code>rev</code>	Reverse lines of a file
<code>rm</code>	Remove files
<code>rmdir</code>	Remove folder(s)
<code>rmmod</code>	Simple program to remove a module from the Linux Kernel
<code>rsync</code>	Remote file copy (synchronize file trees)
<code>screen</code>	Multiplex terminal, run remote shells via ssh
<code>scp</code>	Secure copy (remote file copy)
<code>sdiff</code>	Merge two files interactively
<code>sed</code>	Stream editor
<code>select</code>	Accept user choices via keyboard input
<code>seq</code>	Print numeric sequences
<code>set •</code>	Manipulate shell variables and functions
<code>setfacl</code>	Set file access control lists.
<code>sftp</code>	Secure file transfer program
<code>sha256sum</code>	Compute and check SHA256 (256-bit) checksums
<code>shift •</code>	Shift positional parameters
<code>shopt •</code>	Shell options
<code>shuf</code>	Generate random permutations
<code>shutdown</code>	Shutdown or restart Linux
<code>sleep</code>	Delay for a specified time
<code>slocate</code>	Find files
<code>sntp</code>	Simple Network Time Protocol client program
<code>sort</code>	Sort text files
<code>source •</code>	Run commands from a file '.'
<code>split</code>	Split a file into fixed-size pieces
<code>ss</code>	Socket statistics
<code>ssh</code>	Secure shell client (remote login program)
<code>stat</code>	Display file or file system status
<code>strace</code>	Trace system calls and signals
<code>su</code>	Substitute user identity
<code>sudo</code>	Execute a command as another user
<code>sum</code>	Print a checksum for a file
<code>suspend •</code>	Suspend execution of this shell
<code>sync</code>	Synchronize data on disk with memory
<code>tabs</code>	Set tabs on a terminal
<code>tac</code>	Concatenate and write files in reverse
<code>tail</code>	Output the last part of a file
<code>tar</code>	Store, list or extract files in an archive
<code>tee</code>	Redirect output to multiple files
<code>test •</code>	Evaluate a conditional expression
<code>time</code>	Measure program running time
<code>timeout</code>	Run a command with a time limit
<code>times •</code>	User and system times

Command	Description
tmux	Terminal multiplexer
touch	Change file timestamps
top	List processes running on the system
tput	Set terminal-dependent capabilities, colour, position
traceroute	Trace route to host
trap •	Execute a command when the shell receives a signal
tr	Translate, squeeze, and/or delete characters
true •	Do nothing, successfully
tsort	Topological sort
tty	Print filename of terminal on stdin
type •	Describe a command
ulimit •	Limit user resources
umask •	Users file creation mask
umount	Unmount a device
unalias •	Remove an alias
uname	Print system information
unexpand	Convert spaces to tabs
uniq	Uniquify files
units	Convert units from one scale to another
unix2dos	UNIX to Windows or MAC text file format converter
unrar	Extract files from a rar archive
unset •	Remove variable or function names
unshar	Unpack shell archive scripts
until	Execute commands (until error)
uptime	Show uptime
useradd	Create new user account
userdel	Delete a user account
usermod	Modify user account
users	List users currently logged in
uuencode	Encode a binary file
uudecode	Decode a file created by uuencode
v	Verbosely list directory contents ('ls -l -b')
vdir	Verbosely list directory contents ('ls -l -b')
vi	Text editor
vmstat	Report virtual memory statistics
w	Show who is logged on and what they are doing
wait •	Wait for a process to complete
watch	Execute/display a program periodically
wc	Print byte, word, and line counts
whereis	Search the user's \$path, man pages and source files for a program
which	Search the user's \$path for a program file
while	Execute commands
who	Print all usernames currently logged in
whoami	Print the current user id and name ('id -un')
wget	Retrieve web pages or files via HTTP, HTTPS or FTP
wl-copy	Copy to clipboard (Wayland)
wl-paste	Paste from clipboard (Wayland)
write	Send a message to another user
xargs	Execute utility, passing constructed argument list(s)
xclip	Copy to clipboard (X11)

Command	Description
<code>xdg-open</code>	Open a file or URL in the user's preferred application.
<code>xxd</code>	Make a hexdump or do the reverse
<code>xz</code>	Compress or decompress .xz and .lzma files
<code>yes</code>	Print a string until interrupted
<code>zip</code>	Package and compress (archive) files
<code>.</code>	Run a command script in the current shell
<code>!!</code>	Run the last command again
<code>#</code>	Comment / remark